

Paper 27.25 - Observer Delay Toolkit Supplement

CQE-paper-27.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-27.25.md

Paper 27.25 - Observer Delay Toolkit Supplement

Purpose

This supplement shows how to reproduce Paper 27's observer-frame receipt. The proof is in Paper 27 and its formal verifier; human observer interpretations remain obligations.

Digital Route

Run:

```
`python production/formal-papers/CQE-paper-27/verify_observer_delay_shared_reality.py`
```

The expected status is ``pass_with_interpretive_obligations``. The verifier checks the static Z4 template, the temporal-period refutation, the shared ``C`` invariant, bounded anneal delay, and the non-promotion of consciousness, collapse, simultaneity, and human-latency claims.

Analog Route

Draw a three-cell strip ``L | C | R``. Put one observer token at the left boundary and one at the right boundary. Both read toward the center. Swap ``L`` and ``R`` to simulate the opposite-boundary observer, then verify that the center bead remains unchanged.

Use four cards for the static Z4 frames. Rotate the cards to compute the four-frame label. Do not use the wheel to predict time. Mark any failed temporal prediction as a black follow-up row.

Hidden-Guess Diagnostic

Training mode should hide the label until after the reviewer chooses one:

- static Z4 frame fact,
- temporal Z4 overclaim,
- shared center fact,
- boundary-side residue,
- bounded anneal delay,
- invalid consciousness promotion,
- invalid measurement-collapse promotion.

The answer key distinguishes finite chart receipt from observer interpretation.